

AMAL NIMMY LAL

M.Sc. Computer Science Student

- Worked with LLMs, RAG Pipelines & Machine Learning Models
- Experience with Fraunhofer IEM & DICE Research Group
- ✉ amallal020799@gmail.com
- 🔗 LinkedIn Profile
- 🐙 GitHub Portfolio
- 📍 Paderborn, Germany



Education

10/2022 – Present	Master of Science - Computer Science <i>Universität Paderborn, Germany</i> Relevant Courses: Explainable AI, Unsupervised Learning (R), Prolog.
08/2017 – 06/2021	Bachelor of Technology in CS & Engineering <i>KCC Institute of Technology and Management, India</i>

Work Experience

01/2025 – Present	Student Assistant <i>Fraunhofer IEM (Institute for Mechatronic Systems Design), Paderborn</i> • Currently contributing to the Automated Airport Luggage Handling project by identifying & optimizing baggage stacking and transport, ensuring damage-free loading. • Previously contributed to the FastGate Project by optimizing object localization algorithms to accurately localize aircraft parts and airport infrastructure.
11/2023 – 10/2024	Student Research Assistant <i>DICE Research Group, Paderborn</i> • Co-authored with Aleksandr Perevalov on the Materials AI Agent (formerly LASS KG) project, developing a semantic search interface for Springer Nature. • Fine-tuned BERT variants and LLMs (Mistral, Zephyr) for NER, achieving 90% accuracy, which contributed to the project's successful publication.
01/2022 – 02/2022	Data Science & ML Scholar Intern <i>Tathastu Technologies Pvt. Ltd, Noida, India</i> • Performed exploratory data analysis (EDA) on datasets and engineered novel features, boosting model accuracy. • Applied ML techniques to solve real-world problems across 5 projects.

Achievements & Publication

Publication	CLASS MATE: Cross-Lingual Semantic Search for Material Science Driven by Knowledge Graphs Published in the proceedings of the European Semantic Web Conference (ESWC) 2024, Springer.
Winner	Vibathon Paderborn 2025 (GDG DevFest) Project: Developed an AI Calling Agent to assist English-speaking expats in booking appointments with German doctors, bridging the language barrier. Tech: Implemented end-to-end voice automation with real-time calendar integration. - Secured 1st place among 13 teams for rapid prototyping and generative AI integration.
Winner	Makeathon XCHANGE4INDUSTRY 2025 Project: Developed a "Digital Data Vault" concept for Werkzeugbau Berger GmbH to solve industry-specific data networking challenges. Award: Won 3,000 Euro prize and 1st place against 7 teams (Judged by Fraunhofer IEM & Industry Leaders). - Recognized for combining technical coding skills with clear data management and industry applicability.

Academic Projects

Gen AI	Requirements Generation - Collaborated with a team of five using GPT-4 and RAG, reducing token usage by 15% to optimize project costs. - Spearheaded frontend development with React.js and Streamlit.
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Skills & Interests

AI & ML	&	NLP, Training & Evaluating Models, Explainable AI, Unsupervised Learning, RAG
Languages Frameworks		Python, R, ReactJS, FastAPI, Streamlit, Selenium, Prolog
Interests		Fabric Painting, Bollywood Dancing, Travel & Event Planning